

MOLECULAR GENETICS

Module map

- Module 07: Molecular Genetics
- Connected lab: lab-07_molecular-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 07

Molecular Genetics

1

DNA
structure

2

Replication

3

Transcription

4

Translation

5

Mutation
and gene
expression
evidence**Lab evidence**

lab-07_molecular-genetics.md

MOLECULAR GENETICS

Learning objectives

- Describe DNA structure and base-pairing rules.
- Explain why DNA replication is semiconservative.
- Compare transcription and translation.
- Use codons to connect mRNA sequence to amino acid sequence.
- Explain how mutation can alter protein and trait outcomes.

OBJECTIVE 1

Describe DNA structure and base-pairing rules.

OBJECTIVE 2

Explain why DNA replication is semiconservative.

OBJECTIVE 3

Compare transcription and translation.

OBJECTIVE 4

Use codons to connect mRNA sequence to amino acid sequence.

OBJECTIVE 5

Explain how mutation can alter protein and trait outcomes.

MOLECULAR GENETICS

Topic sequence

- DNA structure: DNA stores information in nucleotide sequences and complementary base pairing.
- Replication: Replication copies DNA using each strand as a template.
- Transcription: Transcription produces RNA from a DNA gene.
- Translation: Translation uses mRNA codons and ribosomes to build proteins.
- Mutation and gene expression evidence: Mutations and expression patterns connect molecular changes to traits.

01

DNA structure

DNA stores information in nucleotide sequences and complementary base pairing.

02

Replication

Replication copies DNA using each strand as a template.

03

Transcription

Transcription produces RNA from a DNA gene.

04

Translation

Translation uses mRNA codons and ribosomes to build proteins.

05

Mutation and gene expression evidence

Mutations and expression patterns connect molecular changes to traits.

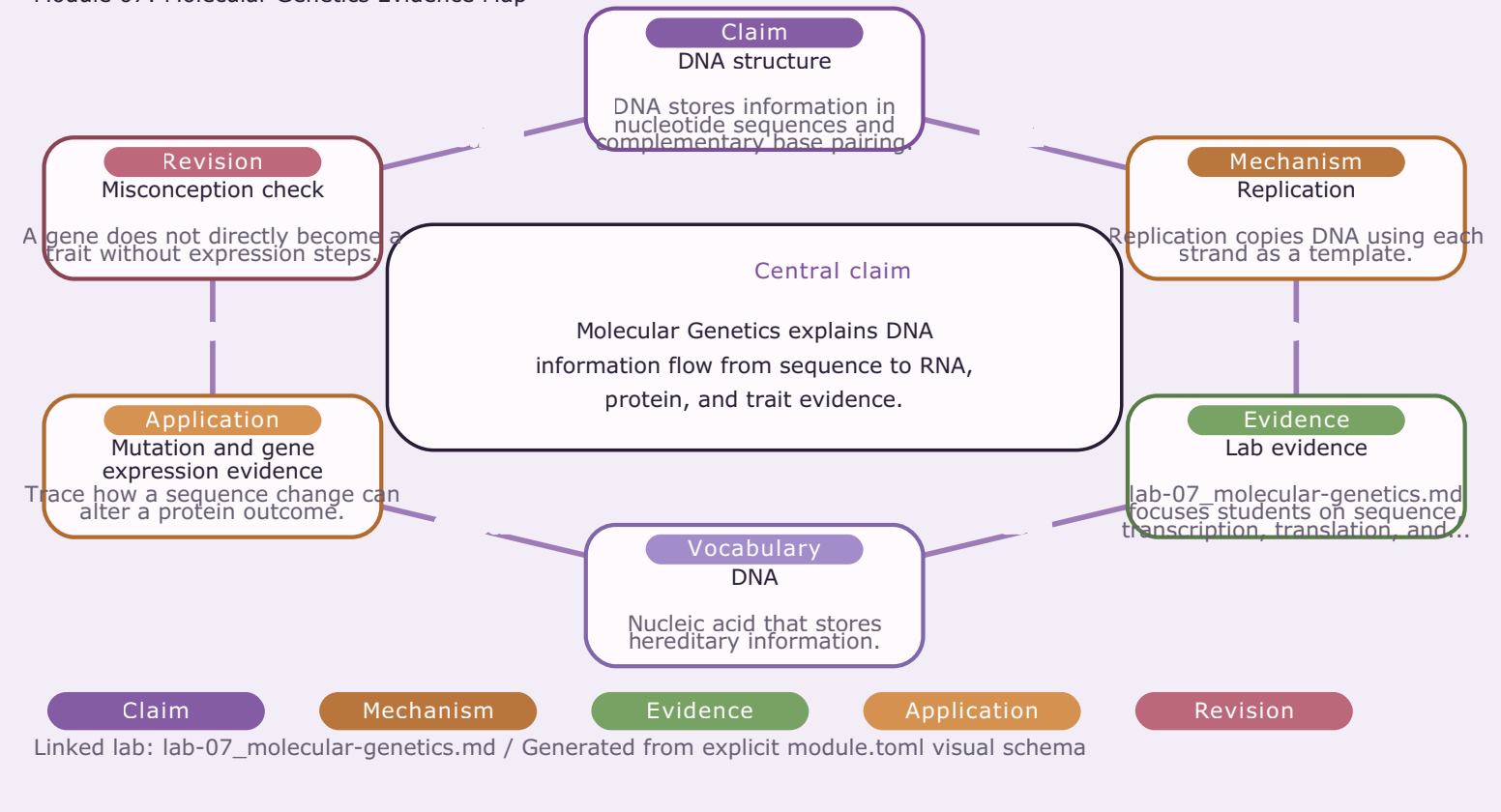
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Module 07: Molecular Genetics Evidence Map

- Module focus: Molecular Genetics
- Central claim: Molecular Genetics explains DNA information flow from sequence to RNA, protein, and trait evidence.
- Key nodes to connect: DNA structure, Replication, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-07_molecular-genetics.md supplies an evidence surface for the map.

Module 07 / Concept Map

Module 07: Molecular Genetics Evidence Map



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Module 07: Molecular Genetics Reasoning Sequence

- Module focus: Molecular Genetics
- Trace the model: DNA structure -> Replication -> Transcription -> Translation
- Inputs become outputs: DNA structure, Replication -> Describe DNA structure and base-pairing rules., Explain why DNA replication is semiconservative.
- Feedback check: lab-07_molecular-genetics.md evidence checks whether students can explain transcription.
- Lab connection: lab-07_molecular-genetics.md gives students a process to observe or test.

Module 07 / Process Model

Module 07: Molecular Genetics Reasoning Sequence

Inputs

- DNA structure
- Replication
- Evidence from lab-07_molecular-genetics.md

Outputs

- Describe DNA structure and base-pairing rules.
- Explain why DNA replication is semiconservative.
- Explain how mutation can alter protein and trait outcomes.

Reasoning sequence



Feedback

lab-07_molecular-genetics.md evidence checks whether students can explain transcription.; Retrieval questions reveal whether terms are...

Constraint

A gene does not directly become a trait without expression steps.; Mutations and expression patterns connect molecular changes to traits.

Linked lab: lab-07_molecular-genetics.md

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Terms as evidence handles

- DNA: Nucleic acid that stores hereditary information.
- Nucleotide: DNA or RNA subunit with sugar, phosphate, and base.
- Replication: Process that copies DNA.
- Transcription: Synthesis of RNA from a DNA template.
- Translation: Synthesis of protein from mRNA codons.
- Codon: Three-base mRNA sequence read during translation.

DNA

Nucleic acid that stores hereditary information.

NUCLEOTIDE

DNA or RNA subunit with sugar, phosphate, and base.

REPLICATION

Process that copies DNA.

TRANSCRIPTION

Synthesis of RNA from a DNA template.

TRANSLATION

Synthesis of protein from mRNA codons.

CODON

Three-base mRNA sequence read during translation.

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Lab connection

- Lab file: lab-07_molecular-genetics.md
- Lab evidence should test or illustrate: Transcription
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION

DNA structure

EVIDENCE

lab-07_molecular-genetics.md

CLAIM

Describe DNA structure and base-pairing rules.

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Contrast check

- Do not stop at: DNA stores information in nucleotide sequences and complementary base pairing.
- Stronger explanation: Mutations and expression patterns connect molecular changes to traits.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

DNA stores information in nucleotide sequences and complementary base pairing.

DEEPER

Mutations and expression patterns connect molecular changes to traits.

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Module 07: Molecular Genetics Retrieval and Lab Check

- Module focus: Molecular Genetics
- Retrieval prompt: How does DNA structure store information?
- Answer check: Describe DNA structure and base-pairing rules.
- Required terms: DNA, Nucleotide, Replication
- Lab connection: lab-07_molecular-genetics.md; lab-07_molecular-genetics.md supplies evidence for sequence, transcription, translation, and gene-expression evidence.

Module 07 / Retrieval Card

Module 07: Molecular Genetics Retrieval
and Lab Check

Routine

Cover notes -> answer aloud
-> cite evidence -> revise

1 Cover notes

2 Answer aloud

3 Cite evidence

4 Revise

How does DNA structure store information?

Check: Describe DNA structure and base-pairing rules.

Why does complementary base pairing matter?

Check: Explain why DNA replication is semiconservative.

What does semiconservative replication mean?

Check: Compare transcription and translation.

How is transcription different from translation?

Check: Use codons to connect mRNA sequence to amino acid sequence.

Terms to use

DNA, Nucleotide, Replication, Transcription, Translation

Lab connection

lab-07_molecular-genetics.md supplies evidence for sequence, transcription, translation, and gene-expression.

Intro biology routine: claim, evidence, reasoning, revision.

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Practice quiz bridge

- 1. In DNA, the base adenine always pairs with:
- 2. DNA replication is described as "semiconservative" because each new double helix contains:
- 3. Which sequence shows the normal flow of genetic information?
- 4. A mutation changes a single DNA base. Why might this matter?

QUESTION 1**Answer first**

In DNA, the base adenine always pairs with:

QUESTION 2**Answer first**

DNA replication is described as "semiconservative" because each new double helix contains:

QUESTION 3**Answer first**

Which sequence shows the normal flow of genetic information?

QUESTION 4**Answer first**

A mutation changes a single DNA base. Why might this matter?

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Synthesis and exit ticket

- Exit claim: explain dna structure using evidence from lab-07_molecular-genetics.md.
- Must include: DNA, Nucleotide, Replication.
- Revision prompt: what evidence would change your explanation?

CLAIM

DNA structure

EVIDENCE

lab-07_molecular-genetics.md

REVISION

What would change your mind?